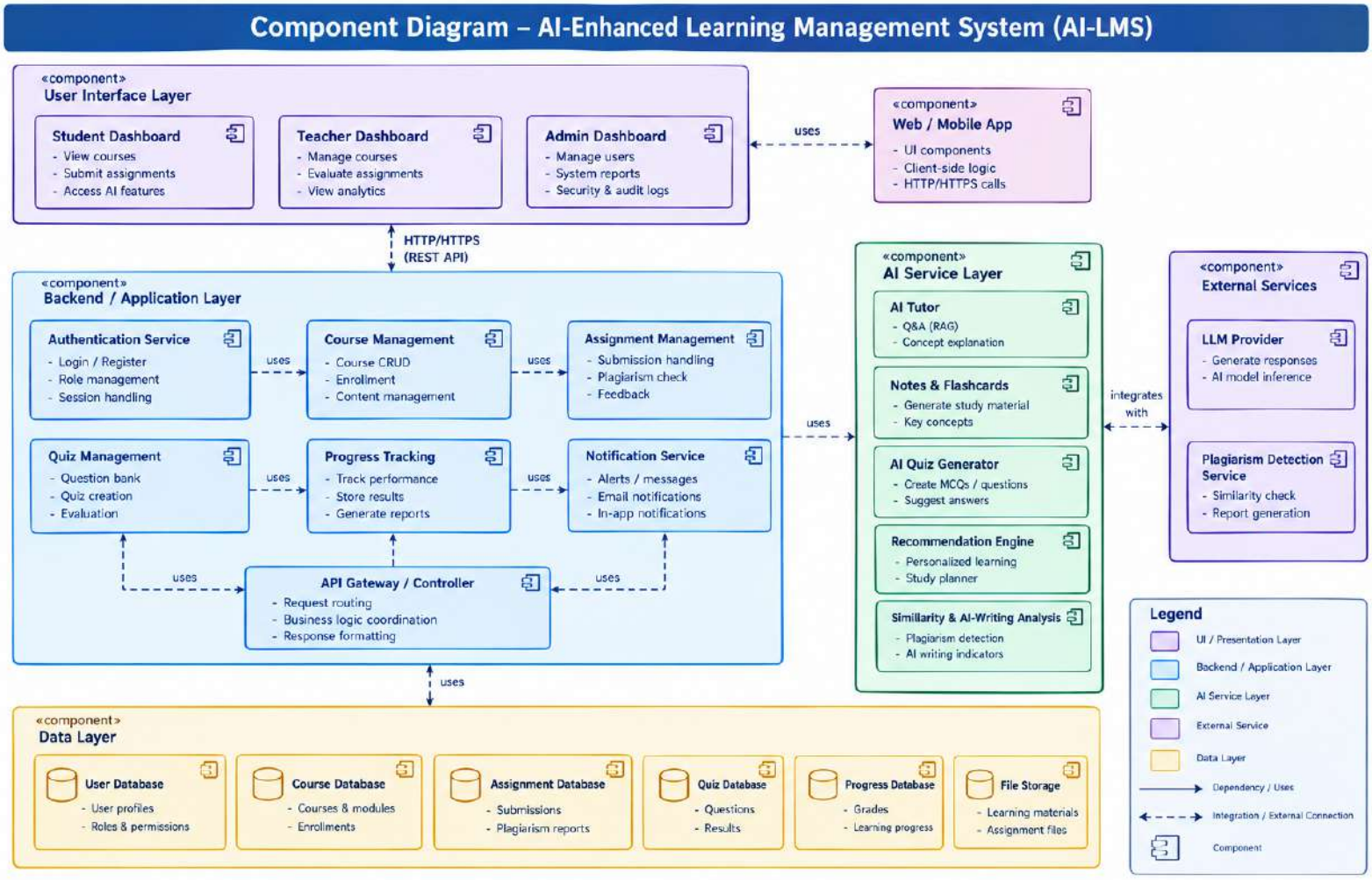

Component Diagram – AI-Enhanced Learning Management System (AI-LMS)
«component»
User Interface Layer
Student Dashboard
- View courses
- Submit assignments
- Access AI features
Teacher Dashboard
- Manage courses
- Evaluate assignments
- View analytics
Admin Dashboard
- Manage users
- System reports
- Security & audit logs
uses
«component»
Web / Mobile App
- UI components
- Client-side logic
- HTTP/HTTPS calls
HTTP/HTTPS
(REST API)
«component»
Backend / Application Layer
Authentication Service
- Login / Register
- Role management
- Session handling
uses
Course Management
- Course CRUD
- Enrollment
- Content management
uses
Assignment Management
- Submission handling
- Plagiarism check
- Feedback
Quiz Management
- Question bank
- Quiz creation
- Evaluation
uses
Progress Tracking
- Track performance
- Store results
- Generate reports
uses
Notification Service
- Alerts / messages
- Email notifications
- In-app notifications
uses
uses
API Gateway / Controller
- Request routing
- Business logic coordination
- Response formatting
uses
«component»
AI Service Layer
AI Tutor
- Q&A (RAG)
- Concept explanation
Notes & Flashcards
- Generate study material
- Key concepts
AI Quiz Generator
- Create MCQs / questions
- Suggest answers
Recommendation Engine
- Personalized learning
- Study planner
Similarity & AI-Writing Analysis
- Plagiarism detection
- AI writing indicators
integrates
with
«component»
External Services
LLM Provider
- Generate responses
- AI model inference
Plagiarism Detection
Service
- Similarity check
- Report generation
«component»
Data Layer
User Database
- User profiles
- Roles & permissions
Course Database
- Courses & modules
- Enrollments
Assignment Database
- Submissions
- Plagiarism reports
Quiz Database
- Questions
- Results
Progress Database
- Grades
- Learning progress
File Storage
- Learning materials
- Assignment files
Legend
UI / Presentation Layer
Backend / Application Layer
AI Service Layer
External Service
Data Layer
Dependency / Uses
Integration / External Connection
Component